

Trinomial multivariate de Vandermonde

(1)

$$V_{n,2} \text{ Vandermonde } (d_k) = \begin{vmatrix} 1 & d_0 & d_0^2 \\ \vdots & \vdots & \vdots \\ 1 & d_{m-2} & d_{m-2}^2 \end{vmatrix}$$

$$P = (X-d_0) \dots (X-d_{m-2}) = \sum_{k=0}^{m-2} a_k X^k + X^{m-2}$$

$$L_n \leftarrow L_n + \sum_{k=0}^{m-2} a_k L_{n-k}$$

$$a_k \leftarrow a_k + \sum_{k=0}^{m-2} d_k a_k = P(d_k) = \begin{cases} 0 & \text{if } k \leq m-2 \\ \prod_{k=0}^{m-2} (d_{m-2} - d_k) \end{cases}$$

$$V_{n,2} = \begin{vmatrix} 1 & d_0 & d_0^2 \\ \vdots & \vdots & \vdots \\ 1 & d_{m-2} & d_{m-2}^2 \end{vmatrix} = \prod_{k=0}^{m-2} (d_{m-2} - d_k) V_{n-2}$$

$$V_{n,2} = \prod_{l=1}^{m-2} \prod_{k=0}^{l-2} (d_l - d_k)$$

$$= \prod_{0 \leq i < j \leq m-2} (d_j - d_i)$$

Donc pour $(d_i)_{i=0}^{m-2} \mid \forall i, j \in \{0, m-2\} \text{ if } i \neq j \Rightarrow d_i \neq d_j$

$\det(V_n) = V_n \neq 0$ donc V_n est inversible

Considérons $\varphi_k: \mathbb{R}[X] \rightarrow \mathbb{R} \quad k \in \{0, m-2\}$

$$p \mapsto \varphi_k(p) = p(d_k)$$

$$\varphi_k(p+q) = \varphi_k(p) + \varphi_k(q)$$

$$\varphi_k \in \mathcal{L}(\mathbb{R}[X])$$

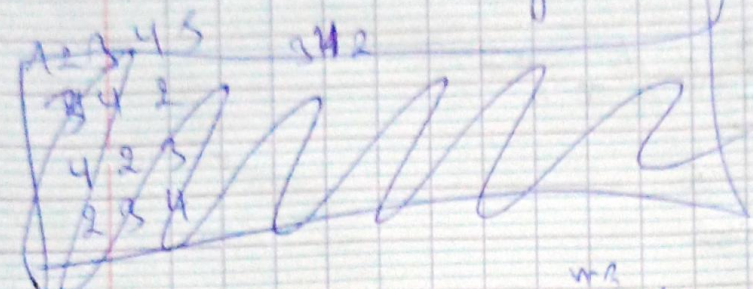
$\{\varphi_k\}_{k=0}^{m-2}$ est une base génératrice car $\bigcap_{k=0}^{m-2} \text{Ker } \varphi_k = \{0\}$

1) Vu que pour $p \in \text{Ker } L_n$ p a n racines distinctes

(2)

Uq, donc et car $p \in \text{Ker } L_n$, $p \in \text{Ker } L_{n-1}$

2) Montrer que $\{L_k\}_{k=0}^{n-2}$ base de $\text{Ker } L_n$



Pour $\forall i \in \{0, n-2\}$,
$$L_i = \prod_{j=0, j \neq i}^{n-2} \frac{(x - \alpha_j)}{\alpha_i - \alpha_j}$$

$\forall k \in \{0, n-2\}$
$$L_k(L_i) = \begin{cases} 1 & \text{si } i=k \\ 0 & \text{sinon} \end{cases}$$

Donc $\{L_i\}_{i=0}^{n-2}$ est la base prduale associe a la base

duale $\mathcal{B} = \{e_k\}_{k=0}^{n-2}$

Il vient
$$\begin{pmatrix} L_0 \\ \vdots \\ L_{n-2} \end{pmatrix} = \begin{pmatrix} 1 & \dots & 1 \\ \vdots & & \vdots \\ 1 & \dots & 1 \end{pmatrix} = L_P^{-1}$$

soit
$$L_i = \frac{1}{\prod_{j=0, j \neq i}^{n-2} (\alpha_i - \alpha_j)} \left(\sum_{k=0}^{n-2} x^k \binom{n-1-k}{k} \sum_{\substack{J \subseteq I_{n-1} \\ |J|=n-1-k}} \prod_{j \in J} \alpha_j \right)$$
 où $I_{n-1} = \{0, n-2\}$

Def 1: $\uparrow$ $\frac{1}{\prod_{k=0}^{n-1} (x - d_k)}$ $\left[\begin{matrix} \sum_{k=0}^{n-1} \prod_{j \neq k} \frac{1}{d_j - d_k} \end{matrix} \right]_{i,j=0}^{n-1}$

Def 2: $\left[\begin{matrix} 1 & \dots & 1 \\ d_0 & \dots & d_{n-1} \end{matrix} \right]_{i,j=0}^{n-1} = \left[\begin{matrix} \frac{1}{\prod_{k=0, k \neq i}^{n-1} (d_i - d_k)} \sum_{k=0, k \neq i}^{n-1} \prod_{j \neq k} \frac{1}{d_j - d_k} \end{matrix} \right]_{i,j=0}^{n-1}$

or $\text{Inv}_1 = \text{Inv}_2$

$= \left[\begin{matrix} \frac{1}{\prod_{k=0, k \neq i}^{n-1} (d_i - d_k)} \sum_{k=0, k \neq i}^{n-1} \prod_{j \neq k} \frac{1}{d_j - d_k} \end{matrix} \right]_{i,j=0}^{n-1}$

or $\text{Inv}_1 = \text{Inv}_2$

Ex 1: $[1]_1^2 = [1]$

$\cdot \left[\begin{matrix} 1 & 1 \\ d_0 & d_1 \end{matrix} \right]_1^2 = \left[\begin{matrix} \frac{d_1}{d_1 d_0} & \frac{-1}{d_1 - d_0} \\ \frac{+d_0}{d_0 - d_1} & \frac{-1}{d_0 - d_1} \end{matrix} \right] = \frac{1}{d_1 d_0} \left[\begin{matrix} d_1 & -1 \\ -d_0 & 1 \end{matrix} \right]$

$\cdot \left[\begin{matrix} 1 & 1 & 1 \\ d_0 & d_1 & d_2 \\ d_0^2 & d_1^2 & d_2^2 \end{matrix} \right]_1^3 = \left[\begin{matrix} \frac{d_1 d_2}{(d_1 - d_0)(d_2 - d_0)} & -\frac{d_1 + d_2}{(d_1 - d_0)(d_2 - d_0)} & \frac{1}{(d_1 - d_0)(d_2 - d_0)} \\ \frac{d_0 d_2}{(d_0 - d_1)(d_2 - d_1)} & -\frac{d_0 + d_2}{(d_0 - d_1)(d_2 - d_1)} & \frac{1}{(d_0 - d_1)(d_2 - d_1)} \\ \frac{d_0 d_1}{(d_0 - d_2)(d_1 - d_2)} & -\frac{d_0 + d_1}{(d_0 - d_2)(d_1 - d_2)} & \frac{1}{(d_0 - d_2)(d_1 - d_2)} \end{matrix} \right]$